Computational Neuroscience — Week 5

Computing in Carbon

Membranes, Ion Channels, and Where Linearity Breaks

These notes cover lectures 1–4 — turning a neuron into a circuit diagram, finding where its behaviour stops being linear, building the Hodgkin–Huxley model that explains the spike, stripping it back to the simplest models that still behave like neurons, and finally putting the geometry back in. Green boxes give the plain-words version of each idea before the mathematics; rust boxes flag the places where the obvious reading is the wrong one.

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Part I

Neuroelectronics

Lecture 1 — membranes, batteries, ion channels, and the first sign of excitability.

1. Zooming In

Weeks 2 to 4 treated neurons largely as black boxes: something that turns a stimulus into spikes, to be characterised by filters, decoders and information rates. This week opens the box.

[Lecture slide omitted from the public copy.] The plan for the week: the electronics of membranes and channels, then simplified models of excitability, then how a neuron's physical shape affects what it computes.

Watch out — the mathematics changes here

Earlier weeks ran on linear algebra and probability. This one runs on **differential equations** — first-order ones, almost throughout. If the earlier material felt like statistics, this feels like physics, because it is.

The goal for this lecture is concrete: take a neuron — soma, dendrites collecting input, axon carrying spikes away — and turn it into a circuit diagram that can be solved.

[Lecture slide omitted from the public copy.] The translation. A cell body with dendrites and a myelinated axon becomes an equivalent circuit of capacitors, conductances and batteries.

2. Circuit Basics

A short refresher, since everything later is built from four rules.

[Lecture slide omitted from the public copy.] The rules of the game.

The four elements

Wire — current flows freely, and the potential is the same all along it. All voltage changes happen *at components*, never along wires.
Capacitor — an insulator that accumulates charge on either face.
Resistor — resists current, like a narrowing in a pipe slowing the flow of water.
Battery — causes a fixed potential drop.

Two laws do the work:

- **Kirchhoff's law** — at any junction of wires the total current is zero. Whatever flows in must flow out.
- **Ohm's law** — the voltage drop across a resistor is $V = IR$. Neuroscience usually prefers the **conductance** $g = 1/R$, so that $I = Vg$.

The idea in plain words

Conductance rather than resistance is not just a notational preference. An ion channel is a hole that is either open or shut, and it is far more natural to say "this many channels are open, so the conductance is this large" than to quote a resistance. Conductances also simply *add* when channels sit in parallel, which is exactly how a membrane is arranged.

3. The Membrane as an RC Circuit

[Lecture slide omitted from the public copy.] A patch of membrane: two layers of lipid with ion channels embedded in it, each channel a pore that lets particular ions through.

The lipid bilayer is a good insulator with conducting fluid on both sides — which is the definition of a **capacitor**. But it is not a perfect insulator: some charge does leak across. So put a **resistor** in parallel with it.

That is the whole model. A capacitor and a resistor in parallel, driven by an injected current I_{ext}.

4. The Passive Membrane Equation

[Lecture slide omitted from the public copy.] Assembling the equation from the three ingredients.

Take the three facts in turn. Kirchhoff says the currents sum to zero:

$$I_R + I_C + I_{\text{ext}} = 0$$

Ohm gives the resistive current, $I_R = V/R$. The capacitive current needs one step of calculus. Capacitance is defined as $C = Q/V$, so $Q = CV$; differentiate in time, and since C is constant,

$$\frac{dQ}{dt} = C\frac{dV}{dt} \qquad \text{and} \qquad \frac{dQ}{dt} \equiv I_C \qquad \Longrightarrow \qquad I_C = C\frac{dV}{dt}$$

The idea in plain words

That last step is the one worth pausing on. Current *is* the rate at which charge moves, so the current into a capacitor is just how fast its stored charge is changing — and since stored charge is proportional to voltage, that is how fast the voltage is changing. A capacitor therefore resists *sudden* changes in voltage rather than resisting current as such.

Putting them together:

$$C\frac{dV}{dt} = -\frac{V}{R} + I_{\text{ext}}$$

A first-order linear differential equation in V — linear because V appears only to the first power, first-order because only its first derivative appears.

5. The Cell Has a Battery

That equation would describe a scrap of membrane floating in a uniform solution. A real cell is not in that situation.

[Lecture slide omitted from the public copy.] The same circuit with a battery added in series with the resistor.

[Lecture slide omitted from the public copy.] Why the battery is there. The cell hoards potassium inside while sodium, chloride and calcium are more concentrated outside; pumps maintain the imbalance.

The idea in plain words

Two forces pull in opposite directions. **Osmosis** drives ions down their concentration gradient — potassium wants to leave. But every ion that leaves carries charge, and the resulting electrical imbalance pulls it back. The potential at which those two forces exactly cancel is the equilibrium potential for that ion.

A concentration difference maintained across an insulating membrane is precisely what a battery is.

The Nernst potential

$$E = \frac{k_B T}{zq} \ln \frac{[\text{inside}]}{[\text{outside}]}$$

with k_B the Boltzmann constant, T temperature, q the ionic charge, and z the number of charges on the ion.

What the battery changes

Part of the voltage drop now falls across the battery rather than the resistor, so the resistive current is driven by $V - V_{\text{rest}}$ rather than by V:

$$I_R = \frac{V - V_{\text{rest}}}{R} \qquad \Longrightarrow \qquad C\frac{dV}{dt} = -\frac{(V - V_{\text{rest}})}{R} + I_{\text{ext}}$$

6. How the Membrane Behaves

Multiply both sides through by R and group the constant terms:

$$RC\frac{dV}{dt} = -(V - V_{\text{rest}}) + I_{\text{ext}}R = -V + \big(V_{\text{rest}} + I_{\text{ext}}R\big)$$

which is already in standard form. Comparing term by term, two quantities fall out:

$$\tau\frac{dV}{dt} = -V + V_\infty \qquad \tau = RC$$

- $\tau = RC$ is the **membrane time constant** — how quickly the voltage responds.
- V_∞ is the **steady state**. Set $dV/dt = 0$ and the equation gives $V = V_\infty$ directly: it is where the voltage settles once any change in I_{ext} is long past.

[Lecture slide omitted from the public copy.] The response to a square pulse of injected current. The voltage rises exponentially toward V_∞, then decays exponentially back when the current is switched off.

$$\text{rising:} \quad V(t) = V_\infty\big(1 - e^{-t/\tau}\big) \qquad\qquad \text{falling:} \quad V(t) = V_\infty\, e^{-t/\tau}$$

The idea in plain words

This is the leaky integrator from Week 2, arrived at from the other direction. There it was a filter shape chosen because it looked like what neurons do; here it drops out of the physics of a capacitor and a resistor. The membrane *is* an RC low-pass filter, and $\tau = RC$ is the same time constant that set the width of the exponential kernel.

7. Ion Channels

So far the membrane has had one resistance and one battery. But ions do not cross it indiscriminately — they go through channels, and channels are selective.

[Lecture slide omitted from the public copy.] Channels come in many kinds — voltage dependent, transmitter dependent, calcium dependent, mechanosensitive, heat sensitive. The trace shows current through a single *channel flicking stochastically between open and closed.*

The idea in plain words

That recording is worth dwelling on. A single channel is not a smoothly varying resistor: it is essentially binary, snapping open and shut at random. The smooth conductances used in the equations are averages over very many such channels.
Ohm's law still applies to one channel, $I = Vg$, so the current depends on two things: how far the voltage is from equilibrium, and how open the channel is. Everything the rest of the week does comes from making g depend on V.

8. Each Ion Has Its Own Battery

Each ionic species has its own equilibrium potential, and a conductance for that ion pulls the membrane voltage *toward* that potential.

[Lecture slide omitted from the public copy.] Approximate equilibrium potentials: $E_{\mathrm{Na}} \approx 50\,mV$, $E_{\mathrm{Ca}} \approx 150\,mV$, $E_{\mathrm{K}} \approx -80\,mV$, $E_{\mathrm{Cl}} \approx -60\,mV$, *against a resting potential near* $-65\,mV$.

Watch out — sodium and potassium pull in opposite directions

This is the single most important fact for what follows. E_{Na} is *above* rest, so opening sodium channels **depolarises** the membrane — drives it positive. E_{K} is *below* rest, so opening potassium channels **hyperpolarises** it. Chloride sits near rest and does little on its own.
An action potential is what happens when those two opposed forces are switched on in sequence.

[Lecture slide omitted from the public copy.] The full circuit. Each ion gets its own parallel branch with its own conductance and its own battery.

By Ohm's law, discounting each branch by its own equilibrium potential:

$$I_i \;=\; g_i\,(V - E_i)$$

The idea in plain words

Read the sign. When V sits above E_i, the current is positive and flows in the direction that drags V back down toward E_i; when V is below, it flows the other way. Every branch is pulling the membrane toward its own battery, and the voltage settles wherever those pulls balance — a tug-of-war whose outcome is set by the conductances.

9. What Makes a Neuron Compute?

Everything so far is linear, and that is a problem.

Watch out — a linear transformation does not compute anything

A linear map just re-represents its input in another coordinate system. It is invertible — run it backwards and you recover exactly what you started with, because nothing was discarded. Computing, arguably, requires the opposite: doing something *irreversible*. Selecting, thresholding, throwing information away. In the Week 2 coding models it was the nonlinearity, not the filter, that made the transformation non-invertible.

[Lecture slide omitted from the public copy.] Responses to injected current pulses of various signs and amplitudes. For most inputs the traces are neat scaled copies of one another, exactly as a linear system requires — until, for certain currents, something entirely different happens.

The idea in plain words

The passive membrane is a linear filter, so on its own it re-represents its input and computes nothing. The interesting behaviour appears only when the responses stop scaling — when a slightly larger input produces something qualitatively different rather than proportionally bigger.
That departure is **excitability**, and it is where the next section picks up.

10. Where This Leaves Us

- A patch of membrane is a **capacitor** (the lipid bilayer) in parallel with a **resistor** (leak through channels), giving $C\,dV/dt = -V/R + I_{\text{ext}}$.
- Ionic concentration gradients, maintained by pumps, act as **batteries**. The **Nernst potential** is where osmotic and electrostatic forces balance for a given ion.
- With the battery included the equation becomes $\tau\,dV/dt = -V + V_\infty$ with $\tau = RC$ — an exponential approach to steady state, which is the leaky integrator of Week 2 derived from physics rather than assumed.
- Each ion has its own conductance and battery, contributing $I_i = g_i(V - E_i)$. Sodium depolarises, potassium hyperpolarises.
- All of this is **linear**, and therefore invertible, and therefore not yet computation. What breaks the linearity is the subject of the rest of the week.

To add from lectures: Next: excitability and the Hodgkin–Huxley model — what happens when the conductances g_i are themselves allowed to depend on the voltage. A supplementary handout, `Solving the RC circuit equations.pdf`, works through the differential equation in full.

Part II

Excitability and the Hodgkin–Huxley Model

Lecture 2 — where the nonlinearity comes from, and how it assembles into a spike.

11. The Missing Nonlinearity

Lecture 1 ended on a puzzle: the membrane model was linear, and linear systems do not compute. Reassemble the circuit with one branch per ion — sodium, potassium, and a non-specific leak g_L for everything else.

[Lecture slide omitted from the public copy.] The equivalent circuit with three parallel branches. The arrow through g_{Na} is the symbol for a variable *resistor.*

Watch out — fixed conductances would change nothing

If g_{Na}, g_{K} and g_L were constants, this circuit would still be linear — merely a more elaborate RC circuit with several batteries. Adding branches buys nothing on its own.
The entire interesting behaviour comes from one fact: **the conductances depend on the voltage**. That is what the variable-resistor arrows mean, and it is the nonlinearity the last lecture went looking for.

12. Inside the Potassium Channel

[Lecture slide omitted from the public copy.] A potassium channel: a selectivity filter, an aqueous pore, a ***gate*** *that blocks it, and a* ***voltage sensor*** *that controls the gate's configuration. Open probability rises when the membrane is depolarised.*

The gate is built from **four subunits**, all of which must be in the right configuration for ions to pass. Writing n for the probability that one subunit is open, and assuming the subunits act *independently*:

$$P_K \;\sim\; n^4$$

[Lecture slide omitted from the public copy.] Each subunit flickers open and closed on its own. Current flows only in the moments when all four happen to be open at once — which is what the single-channel recording below shows.

The idea in plain words

Independence is doing real work here. If the four subunits were coupled, the channel's open probability would not be a simple product, and none of the algebra that follows would go through. That n^4 is an *assumption* about molecular structure, not something read off the data — and, as the end of the lecture notes, a remarkably good guess.

13. The Kinetics of One Gate

A subunit has two states: open with probability n, closed with probability $1 - n$. Transitions between them happen at rates that themselves depend on voltage — $\alpha_n(V)$ for closed→open, $\beta_n(V)$ for open→closed.

[Lecture slide omitted from the public copy.] The two-state picture and its rate equation.

$$\frac{dn}{dt} \;=\; \underbrace{\alpha_n(V)\,(1-n)}_{\text{gained from closed}} \;-\; \underbrace{\beta_n(V)\,n}_{\text{lost from open}}$$

The idea in plain words

Read it as bookkeeping on a population. The open fraction grows in proportion to how much is currently closed (times the rate of opening), and shrinks in proportion to how much is currently open (times the rate of closing). Nothing more than that.

This is the same shape of equation as the passive membrane, so it can be rearranged the same way:

[Lecture slide omitted from the public copy.] The rewrite into time-constant and steady-state form.

$$\tau_n(V)\frac{dn}{dt} = n_\infty(V) - n, \qquad \tau_n(V) = \frac{1}{\alpha_n + \beta_n}, \qquad n_\infty(V) = \frac{\alpha_n}{\alpha_n + \beta_n}$$

The idea in plain words

Exactly the structure of $\tau\, dV/dt = -V + V_\infty$ from lecture 1, and it says the same thing: n relaxes exponentially toward a target n_∞ at a speed set by τ_n. The difference is that here *both* the target and the speed depend on the voltage — which is where the whole coupling between voltage and conductance enters.

14. Sodium: Activation and Inactivation

The sodium channel works the same way, with one crucial addition.

[Lecture slide omitted from the public copy.] Sodium has three activation subunits and *a separate inactivation gate — a ball-and-socket mechanism that can block the pore even when the activation gates are open.*

Write m for the probability that one activation subunit is open, and h for the probability that the blocking gate is *not* in place. Then

$$P_{\mathrm{Na}} \sim m^3 h$$

so the channel passes current only when three subunits are open *and* the inactivation gate is out of the way.

Watch out — voltage moves m and h in opposite directions

Depolarisation *increases* m — opening the activation gates — but *decreases* h, swinging the blocking gate into place. There is therefore only a window of voltage in which sodium can actually flow.

This is why sodium currents are **transient**, or self-limiting. As soon as sodium flows, it drags the membrane toward E_{Na}; that rising voltage drives h toward zero, which shuts the channel again. The current switches itself off, and that is one of the mechanisms that terminates a spike.

15. Assembling the Model

Three gating variables, each with its own rate equation of identical form:

[Lecture slide omitted from the public copy.] The three gating equations, and the voltage-dependent conductances they produce.

$$g_K(V) = \bar{g}_K\, n^4 \qquad\qquad g_{\mathrm{Na}}(V) = \bar{g}_{\mathrm{Na}}\, m^3 h$$

where $\bar{g}$ is the maximum conductance if every channel were open.

[Lecture slide omitted from the public copy.] Kirchhoff and Ohm applied to the full circuit: capacitative current on the left, ionic currents and injected current on the right.

$$C_m \frac{dV}{dt} \;=\; -\sum_i g_i\,(V - E_i) \;+\; I_e$$

16. The Hodgkin–Huxley Equations

[Lecture slide omitted from the public copy.] The complete system — one equation for the voltage, three for the gating variables.

$$\begin{aligned}
-C_m \frac{dV}{dt} &= g_L(V - E_L) + \bar{g}_K n^4 (V - E_K) + \bar{g}_{\mathrm{Na}} m^3 h\,(V - E_{\mathrm{Na}}) \;-\; I_e \\
\frac{dn}{dt} &= \alpha_n(V)(1 - n) - \beta_n(V)\, n \\
\frac{dm}{dt} &= \alpha_m(V)(1 - m) - \beta_m(V)\, m \\
\frac{dh}{dt} &= \alpha_h(V)(1 - h) - \beta_h(V)\, h
\end{aligned}$$

The idea in plain words

Four coupled first-order differential equations, and that is the whole model. The voltage equation is driven by the gating variables through the conductances; the gating equations are driven by the voltage through α and β. That two-way coupling is the entire source of the dynamics.

17. Anatomy of a Spike

Two plots explain everything the model does.

[Lecture slide omitted from the public copy.] Left: the steady states — where each variable is heading *at a given voltage. Right: the time constants* — how fast it gets there.

Read off the two facts that matter:

- **Steady states.** n_∞ and m_∞ rise with voltage; h_∞ *falls.* All are sigmoidal.
- **Time constants.** τ_m is much the smallest. Sodium activation is the fastest thing in the system; h and n are slower, and on a similar scale to each other.

The idea in plain words

That separation of timescales is the whole trick. All three variables are chasing their steady states, but m gets there first and the other two lag behind. A spike is what a system does when a fast positive feedback is caught up by slower negative ones.

The sequence

[Lecture slide omitted from the public copy.] The steady states and time constants alongside the resulting time course of V, m, h and n through a single spike.

1. **Rest, then a nudge.** Sitting near $-60\,\text{mV}$, some input depolarises the membrane.
2. **Runaway positive feedback.** Because τ_m is shortest, m reacts first and sodium channels open. Sodium flows in, pulling V toward E_{Na}, which raises m further, which opens more channels. The rise is explosive.
3. **Inactivation catches up.** With a delay, h finally approaches its steady state — and at these raised voltages h_∞ is heading toward zero. Since sodium needs m^3h, the channels shut.
4. **Potassium catches up too.** On a similar timescale n rises, opening potassium channels and dragging V down toward $E_K \approx -80\,\text{mV}$.
5. **Back to rest.** The two slow processes together — the lecture's *double whammy* — return the membrane to baseline.

The idea in plain words

Both slow processes push the same way. Sodium inactivating removes the depolarising drive, and potassium activating adds a hyperpolarising one. That is why the falling phase is so decisive, and why the membrane briefly overshoots below rest before settling.

18. What Hodgkin and Huxley Actually Did

Watch out — they had no knowledge of ion channels

All of this was inferred in the early 1950s, before anyone knew that ion channels existed — let alone that they were built from subunits. The model was reverse-engineered from voltage-clamp measurements of currents.

Notice how little of the model is exotic. The circuit is linear; the gating equations are simple first-order rate equations. Only *two* ingredients are not:

- the multiplicative factors n^4 and m^3h that link subunit behaviour to channel conductance;
- the voltage dependence of the rates $\alpha(V)$ and $\beta(V)$.

The idea in plain words

Those exponents were fitted to make the curves match. That they later turned out to correspond to a real molecular architecture — four potassium subunits, three sodium activation gates plus a separate inactivation mechanism — is one of the more striking cases of

a phenomenological model turning out to be structurally true.

19. Two Directions From Here

[Lecture slide omitted from the public copy.] Hodgkin–Huxley as a fork in the road.

- **Toward biophysical realism.** Dig into channel physics, the signalling cascades that modulate it, and the hundreds of channel types — calcium-dependent, chloride, mixed — operating on very different timescales. That diversity is what makes different neuron types suited to different roles. And real neurons are not patches of membrane but extended structures, so **geometry** enters too.
- **Toward simplification.** Write down reduced models that keep the essential dynamics but are analytically tractable, or at least fast enough for large-scale simulation while still respecting the biophysics.

20. Where This Leaves Us

- Excitability comes from **voltage-dependent conductances**. With fixed conductances the circuit stays linear no matter how many branches it has.
- A gate's open fraction obeys $\tau(V)\,dx/dt = x_\infty(V) - x$ — the same form as the membrane equation, but with both the target and the speed set by voltage.
- Potassium is **persistent**: $g_K = \bar{g}_K n^4$. Sodium is **transient**: $g_{\mathrm{Na}} = \bar{g}_{\mathrm{Na}} m^3 h$, because voltage raises m but lowers h, so it switches itself off.
- The spike is a **separation of timescales**: fast sodium activation gives runaway positive feedback, then slower inactivation and potassium activation together shut it down.
- The model is four coupled first-order ODEs, almost all of it linear — the nonlinearity lives in n^4, m^3h, and the voltage dependence of the rates.

> *To add from lectures: Next: simplified models that capture these dynamics more cheaply, and then neuronal geometry — how dendrites affect what a neuron computes.*

Part III

Simplified Models

Lecture 3 — reduced models that keep the essential dynamics, from integrate-and-fire to the phase plane.

21. Why Simplify

[Lecture slide omitted from the public copy.] The fork from the end of lecture 2. This part takes the right-hand branch.

Two quite different motives push toward simpler models:

- **Analytical tractability** — a model simple enough to reason about on paper, so that one can understand *why* a given ion channel produces a given behaviour.
- **Computational cost** — a model cheap enough to simulate in very large networks, while still respecting something of the biophysics.

What we are trying to reproduce

[Lecture slide omitted from the public copy.] Real neurons driven by noisy input. A cortical neuron, thalamic neurons at two different depolarisations — note the bursting in one and single spikes in the other — and a motor neuron firing very regularly.

[Lecture slide omitted from the public copy.] Five idealised firing patterns, each suggesting a different kind of code.

The idea in plain words

Work down that figure and each row implies a different reading:
- Two regular trains at different frequencies — a plain *rate code*.
- Same mean rate, spikes shifted slightly — local frequency carrying information, like an FM signal.
- So much timing variability that *precise spike times* may matter.
- Bursts among single spikes — perhaps *two distinct symbols* in the code.

A reduced model earns its keep by being able to produce all of these, not just regular firing.

22. The General Form

Everything in this lecture is a search for one function. Write

$$\frac{dV}{dt} = f(V) + I(t)$$

and set $C = 1$ to avoid carrying constants. The task is to choose $f(V)$.

Starting linear

Away from a spike, a neuron behaves close to linearly — so start there, with $f(V) = -a(V - V_0)$, exactly the passive membrane of lecture 1.

*[Lecture slide omitted from the public copy.] The linear choice. Where $f(V)$ crosses zero is a **fixed point**: $dV/dt = 0$, so V stays put.*

Reading stability off the slope

Below the crossing $f(V) > 0$, so V increases; above it $f(V) < 0$, so V decreases. Both sides flow *inward*, making V_0 a **stable** fixed point.
In one dimension this generalises: **negative slope at a fixed point means stable, positive slope means unstable**.

23. Integrate-and-Fire

A linear equation will never spike on its own, so the spike is added by hand.

[Lecture slide omitted from the public copy.] Three extra ingredients: a threshold V_{th}, a peak $V_{\max}$, and a reset V_{reset}.

The integrate-and-fire neuron

Behaves like a passive membrane, with the added rule that when $V \to V_{\text{th}}$ a spike is recorded and $V \to V_{\text{reset}}$. Here E_L is the resting potential V_0.

[Lecture slide omitted from the public copy.] The model driven by a fluctuating current. The output is hard to tell apart from a real spike train.

Watch out — the spike is pasted on, not produced

Nothing in the equation generates the upstroke — it is a bookkeeping rule applied from outside. The model integrates until a line is crossed, then *declares* a spike. That is enough for many purposes, but it means the model has no intrinsic notion of excitability.

24. Making It Intrinsically Excitable

To do better, give $f(V)$ a region where the voltage can run away on its own.

[Lecture slide omitted from the public copy.] Bending $f(V)$ back up above zero creates a second *crossing.*

The idea in plain words

Now there are two fixed points. The lower one is stable as before. The upper one has *positive* slope, so it is **unstable**: nudge V above it and $f(V)$ is positive, which pushes V higher still, which makes $f(V)$ larger again.
That unstable point *is* the threshold — not a rule imposed from outside, but a property of the dynamics. Cross it and the model spikes by itself.

[Lecture slide omitted from the public copy.] A quadratic $f(V)$ does the job. A maximum and a reset are still needed, since nothing yet stops the runaway.

Exponential integrate-and-fire

[Lecture slide omitted from the public copy.] A choice that fits cortical neurons well: linear below threshold plus an exponential term that mimics the sharp rise of a real spike.

$$f(V) \;=\; -a(V - V_0) \;+\; \exp\!\left(\frac{V - V_{\text{th}}}{\Delta}\right)$$

where Δ sets how sharply the nonlinearity turns on.

25. The Theta Neuron

[Lecture slide omitted from the public copy.] Treating the voltage as a phase θ *around a circle.*

$$\frac{d\theta}{dt} \;=\; 1-\cos\theta \;+\; (1+\cos\theta)\,I(t)$$

The idea in plain words

A spike is declared when θ passes π — and because a phase wraps around to $-\pi$ automatically, the **reset comes for free**. No extra rule is needed. This is provably equivalent to the one-dimensional quadratic model.
It also fires regularly with no input at all, which makes it a natural model for neurons that oscillate on their own.

26. Why One Variable Is Not Enough

The maximum and reset are still inelegant. What if $f(V)$ simply came back down instead?

[Lecture slide omitted from the public copy.] Bending $f(V)$ down again creates a third crossing — with negative slope, so stable.

Watch out — this fixes the runaway and creates a worse problem

The voltage now rises to the third fixed point and *stops* there, which is what we wanted. But being stable, it also *stays* there. The system is **bistable**: it has two resting states and no way back from the upper one.

The idea in plain words

Hodgkin–Huxley already told us what is missing. Two mechanisms returned the voltage to rest — sodium inactivating, and potassium activating — and both are slow processes that respond *to* the voltage rather than being instantaneous functions of it.
A single equation in V cannot express that. It needs a second variable with dynamics of its own.

Add a **recovery variable** u:

$$\frac{dV}{dt} = F(V) + G(u) + I(t), \qquad \frac{du}{dt} = -u + H(V)$$

where H makes u grow when V is large, and G is negative so that a large u drags V back down. That is inactivation, in its cheapest possible form.

27. The Phase Plane

With two variables, the picture is no longer a curve but a plane.

[Lecture slide omitted from the public copy.] The phase plane, spanned by V and u. Green: the V-nullcline, where $dV/dt = 0$. Red: the u-nullcline, where $du/dt = 0$. They cross at the resting state.

Nullclines

The V**-nullcline** is the set of points where $dV/dt = 0$; the u**-nullcline** where $du/dt = 0$. A true **fixed point** is where the two intersect — both derivatives vanish at once.

The idea in plain words

Each nullcline divides the plane in two: on one side of the green curve V is increasing, on the other decreasing, and likewise for u about the red curve. Together they organise the whole flow without solving anything.
A spike is a *loop* through that plane. Pushed right of the green curve, V shoots up fast — the upstroke. Crossing the green curve reverses the direction of V, but u is still rising, so the trajectory sweeps over and back. Crossing the red curve lets u fall at last, and the path returns to rest. Plotted against time, that loop is an action potential.

There is a great deal more here — multiple fixed points, limit cycles, bifurcations as the input changes — easily a course of its own. The lecture points to Izhikevich's book, and to online material by Wulfram Gerstner and Bard Ermentrout.

28. The Simple Model

Zoom in on the region of the phase plane where the two nullclines nearly meet, and keep only what is needed.

[Lecture slide omitted from the public copy.] Izhikevich's simple *model: a quadratic voltage equation with linear coupling to a linear recovery variable.*

$$V' = -\alpha V + \beta V^2 + \gamma - u + I(t), \qquad u' = a\,(bV - u)$$

The idea in plain words

The quadratic is the **minimal nonlinearity that gives excitability** — anything less and the unstable fixed point does not exist. Everything else is linear: u grows when V grows, and subtracting u in the voltage equation supplies the inactivation.
The higher-order voltage dynamics that would let the model recover on its own have been discarded, so a maximum and a reset return — and now u needs resetting too.

[Lecture slide omitted from the public copy.] The model with its four parameters: a the decay rate of u, b its sensitivity to V, and c, d the resets of V and u.

[Lecture slide omitted from the public copy.] What four parameters buy. Regular spiking, intrinsic bursting, chattering, fast spiking, thalamocortical responses, spike-frequency adaptation, low-threshold spiking — and a resonator.

Watch out — resonance is the one integrate-and-fire cannot do

Subthreshold resonance — a tendency to oscillate in response to input without necessarily spiking — requires *two* variables playing against each other. No one-dimensional model, integrate-and-fire included, can produce it.
That is the clearest argument for paying the cost of a second variable: it is not a refinement,

it is a different repertoire.

29. Where This Leaves Us

- Every model here has the form $dV/dt = f(V) + I(t)$. The design question is what shape to give f.
- In one dimension, **stability is read off the slope** — negative stable, positive unstable.
- **Integrate-and-fire** is linear with the spike pasted on by rule. Bending $f(V)$ upward adds an unstable fixed point that acts as an *intrinsic* threshold; quadratic and exponential forms both work, and the **theta neuron** gets its reset free by treating voltage as a phase.
- A third fixed point stops the runaway but makes the model **bistable**. Escaping that requires a second, slower **recovery variable** — the reduced-model stand-in for sodium inactivation and potassium activation.
- With two variables the natural tool is the **phase plane**, where nullclines organise the flow and a spike is a loop.
- Izhikevich's **simple model** keeps a quadratic voltage equation and a linear recovery variable, and reproduces a wide range of real firing patterns from four parameters — including subthreshold resonance, which one-variable models cannot reach.

> *To add from lectures: Next: the other branch of the fork — neuronal geometry, and how dendritic structure affects what a neuron computes.*

Part IV

Dendritic Computation

Lecture 4 — the other branch of the fork: geometry, cable theory, and what dendrites contribute.

30. The Embarrassment of Real Neurons

Everything so far has treated the neuron as a *point* — one patch of membrane, one voltage. Zoom out and that looks increasingly hard to defend.

[Lecture slide omitted from the public copy.] Real dendritic arbors.

[Lecture slide omitted from the public copy.] On the left, what neurons actually look like. On the right, what we have been modelling.

The idea in plain words

The honest position is that nobody yet knows what level of description is required to understand how brains work — and there probably is not a single answer. That is a reason to pursue several levels at once rather than to pick one, which is why this lecture and the previous one head in opposite directions from the same starting point.

31. Geometry Matters

Does the shape actually change anything? Inject current in one place and record in another.

[Lecture slide omitted from the public copy.] Left: inject at the cell body, record in a dendrite. Right: inject in a dendrite, record at the cell body. Note the change of vertical scale between the two — 25 mV against 1 mV.

In both directions the signal arrives **delayed**, **attenuated** and **broadened**. Two further wrinkles:

- A *thinner* dendrite gives a *larger* voltage change for the same injected current.
- But the further away the input, the more it is filtered and attenuated along the way.

The idea in plain words

So a synapse's influence at the soma depends on *where it lands*, not just how strong it is. Two identical synapses, one proximal and one distal, are not interchangeable. That alone means geometry is part of the computation rather than packaging around it.

32. Cable Theory

The theory needed here was worked out by Lord Kelvin in the 1850s, for undersea telegraph cables.

[Lecture slide omitted from the public copy.] A tube of membrane. The walls have the same capacitance and resistance as before, but now there is also the resistance of the cable's interior, and current can flow along *the cable as well as out through it.*

Watch out — the mathematics steps up a level

V is now a function of *both* space and time, $V(x, t)$. That means partial differential equations rather than ordinary ones — the first time in this course the distinction has mattered.

Two resistances now matter: r_m, the membrane resistance of a thin slice, and r_i, the axial resistance along the inside. The external medium is treated as perfectly conducting and ignored.

33. The Cable Equation

Current leaving through the membrane is the i_m of lecture 1. What is new is the current flowing *down* the cable, driven by voltage differences along x and opposed by r_i. Combining the two gives an equation that is second order in space:

[Lecture slide omitted from the public copy.] The cable equation, and its rewriting to expose the natural scales.

$$\lambda^2 \frac{\partial^2 V_m}{\partial x^2} = \tau_m \frac{\partial V_m}{\partial t} + V_m$$

$$\tau_m = r_m c_m \qquad \lambda = \sqrt{r_m / r_i}$$

The idea in plain words

Compare it with diffusion. The second derivative in space is exactly the diffusion term — signals spread out the way heat or perfume does. What makes this different is the extra term linear in V: a leak that removes signal as it spreads.
The rearrangement is the same trick as lecture 1. There, grouping constants revealed a time constant $\tau = RC$. Here it reveals a time constant *and* a **space constant** λ, the natural distance scale of the cable.

Why $\lambda = \sqrt{r_m/r_i}$

What stops a signal travelling is leakage out through the walls. A *higher* membrane resistance r_m keeps current inside the cable, so the signal goes further. A *lower* internal resistance r_i lets it flow more freely along the cable, so again it goes further. The ratio of the two sets the distance scale.

34. How Far Does a Signal Travel?

Imagine being a synapse partway along a dendrite, trying to get a message to the soma.

[Lecture slide omitted from the public copy.] Inject a constant *current at $x = 0$ on an infinite cable. The voltage falls off exponentially with distance: $v(x) \propto e^{-|x|/\lambda}$.*

Watch out — shouting louder does not help

Even with current injected continuously, the voltage change reaches only about one space constant. The decay is set by the cable's properties, not by the size of the input.

[Lecture slide omitted from the public copy.] Now a brief pulse *at $t = 0$, $x = 0$. Left: the profile spreads in space and shrinks with time. Right: what an observer sitting at $x = 0$, 0.5λ, 1λ and 2λ sees.*

The idea in plain words

This looks like diffusion — but with one crucial difference. Spray perfume in a room and it spreads out, yet the total amount of perfume is conserved. Here the *total* voltage signal decays away, because of that extra linear term: the signal is leaking out through the membrane as it spreads.
Read the numbers off the right-hand plot. At one space constant there is still a respectable deflection; by two space constants the pulse is down to roughly **5%** of its original size. Signals do not travel far passively.

Tracking the peak gives a propagation velocity, which is just the ratio of the two natural scales:

$$c = \frac{2\lambda}{\tau}$$

[Lecture slide omitted from the public copy.] The full solution for a pulse input: a Gaussian diffusive spread *multiplied by an* exponential decay *prefactor. Arbitrary inputs follow by decomposing them into pulses and summing.*

35. Compartmental Models

[Lecture slide omitted from the public copy.] Two reasons the analytic route runs out.

- Real dendrites **branch**, repeatedly and irregularly.
- Many dendrites are **active** — they contain ion channels, which makes the problem nonlinear.

The answer is to divide and conquer.

[Lecture slide omitted from the public copy.] Chop the arbor into compartments, each small enough that radius and channel density are roughly constant within it.

The idea in plain words

Inside a compartment there is no spatial dependence left, so each one gets an ordinary dV/dt equation — exactly the kind we already know how to write. The geometry re-enters only through *coupling* between neighbouring compartments.
A partial differential equation in x and t has become a large set of coupled ordinary differential equations in t. That is a trade of elegance for tractability, and it is what makes simulation possible.

Rall's simplification

[Lecture slide omitted from the public copy.] If a branch and its two daughters satisfy a particular diameter relation, the pair can be replaced by a single equivalent cable — and the collapse can be applied repeatedly until the whole tree becomes one cylinder attached to the soma.

The 3/2 power rule

$$d_{11}^{3/2} + d_{12}^{3/2} = d_1^{3/2}$$

When the daughter diameters satisfy this, the two branches are impedance matched to the parent and can be merged into one cable of equivalent surface area and electrotonic length.

The idea in plain words

Remarkably, real dendrites satisfy this approximately — and even where the match is imperfect, the resulting approximation is often quite accurate. It is what makes analytic work on passive trees possible at all.
Its limitation is equally clear: it says nothing about ion channels, which make the problem nonlinear, and channel densities often vary along a dendrite in ways that matter.

The full machinery

[Lecture slide omitted from the public copy.] Each compartment gets an equation like the Hodgkin–Huxley membrane, with its own conductances, capacitance and resistance — plus terms coupling it to its neighbours.

The coupling current into compartment 2 from compartment 1 is $g_{1,2}\,(V_1 - V_2)$, with the coupling conductance set by the area of the connection and whether the compartments straddle a branch point.

Watch out — the coupling conductances are not symmetric

$g_{1,2} \neq g_{2,1}$ in general, which is why each connection carries two values. It follows from the compartments having different geometries, not from any asymmetry in the physics.

Models of this kind, built from microscopic reconstructions of real cells, are collected at **ModelDB**, maintained at Yale.

36. What Dendrites Add

[Lecture slide omitted from the public copy.] A survey of proposed roles — logical operations, coincidence detection, low-pass filtering and attenuation, segregation and amplification. After London and Hausser, 2005.

- **Location sets effective strength**, through the passive filtering above.
- **Synaptic scaling.** In hippocampus, dendrites appear to have *solved* that problem: inputs arrive at the soma with a similar shape regardless of where they entered — the synapses compensate for their own distance.
- **Interaction depends on placement.** Two inputs on separate branches contribute independently; two on the *same* branch can sum sublinearly or superlinearly, the latter giving amplification.
- **Dendrites can spike**, typically calcium spikes, thanks to their own channels.
- **Coincidence and plasticity.** Dendritic spikes together with *back-propagating* action potentials from the soma provide a mechanism for detecting coincidence and driving synaptic plasticity — the subject of the following weeks.

37. Two Computations Built From Cables

Sound localisation

[Lecture slide omitted from the public copy.] The Jeffress model. Signals from the two ears run in opposite directions along tapped delay lines; each neuron fires only when inputs from both sides arrive simultaneously.

The idea in plain words

We localise sound partly from the tiny difference in arrival time at the two ears. Each coincidence detector receives its two inputs after travelling different distances, so it has its own built-in delay. The neuron that fires hardest is the one whose *dendritic* delay exactly cancels the *interaural* delay.
The elegance is in the output format: a difference of tens of microseconds is converted into a **place code**, where *which* neuron fires names the timing difference, and hence the direction of the sound.

Direction selectivity

[Lecture slide omitted from the public copy.] Rall's proposal. Inputs from neighbouring points in space arrive at positions along a dendrite in the same spatial order.

The idea in plain words

Stimulate the *far* end first and the signal starts travelling toward the soma; the next input arrives closer in, joins it, and the growing wave gathers the rest as it goes — reaching the soma large enough to cross threshold.
Stimulate the *near* end first and each early input has decayed away before the later ones arrive. Nothing accumulates.
Same inputs, same total, opposite order — and only one direction produces a spike. The asymmetry comes entirely from the cable's propagation delay.

Watch out — the evidence is mixed

This may not be the full explanation of direction selectivity in retinal ganglion cells. But it makes a testable prediction — that firing should depend on the *sequence* of inputs along the dendrite — and Hausser's lab confirmed it, stimulating synthetic input sequences and finding that scrambled orders were discriminable.

38. Where This Leaves Us

[Lecture slide omitted from the public copy.] Where to read further.

- Voltage in a dendrite depends on space as well as time, so the governing equation is a **partial** differential equation — Kelvin's cable equation, which is diffusion plus a leak.
- Grouping constants gives two natural scales: $\tau_m = r_m c_m$ and the **space constant** $\lambda = \sqrt{r_m / r_i}$, with signals propagating at $c = 2\lambda/\tau$.
- Passive signals decay *fast* — to about 5% within two space constants — so where a synapse sits on the tree materially changes its influence.
- Branching and active channels make analytic solutions hopeless, so real modelling uses **compartments**: one ODE each, coupled to neighbours. **Rall's 3/2 rule** collapses passive trees where it applies.
- Dendrites are not passive cabling. They filter, segregate, amplify, spike on their own, and can implement computations — **delay lines** for sound localisation, and sequence sensitivity for **direction selectivity**.

To add from lectures: This closes the biophysics half of the course. What follows moves up a level again, to how these cellular components are wired together into networks that learn.